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Topic 1 — Key Concepts in Chemistry

Edexcel GCSE (9-1) Chemistry (1CH0) — Paper 1 and Paper 2 (examined on both papers)

Topic 1 is the toolkit for the whole course. It covers how we write chemical formulae and equations, what atoms are made of, how the periodic table is organised, how atoms bond together, how bonding explains the properties of substances, and how to do the core chemical calculations (relative formula mass, empirical formulae, concentration and the mole). Because it is examined on **both** papers, marks from this topic turn up in nearly every question — time spent here pays off everywhere.

Content marked **(HT)** is Higher Tier only. Topic 1 is shared with GCSE Combined Science; the separate-chemistry-only content of 1CH0 is concentrated in Topics 5 and 9.

1.1 Formulae, Equations and Hazards

Formulae you must recall

You are expected to know the formulae of common elements, simple compounds and ions without being given them. Many non-metal elements exist as **diatomic molecules** (two atoms bonded together).

Type	Examples you must know
Diatomic elements	H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2
Monatomic elements	Metals (Na, Mg, Fe, Cu…), carbon (C), the noble gases (He, Ne, Ar)
Simple compounds	H_2O, CO_2, CO, NH_3, CH_4, HCl, H_2SO_4, HNO_3, NaCl, NaOH, $CaCO_3$
Positive ions	H^+, Na^+, K^+, Li^+, Ag^+, NH_4^+ (ammonium), Mg^{2+}, Ca^{2+}, Cu^{2+}, Zn^{2+}, Fe^{2+}, Fe^{3+}, Al^{3+}

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Negative ions	Cl^-, Br^-, I^-, OH^- (hydroxide), NO_3^- (nitrate), O^{2-}, S^{2-}, SO_4^{2-} (sulfate), CO_3^{2-} (carbonate)

Naming ionic compounds: the metal (or ammonium) is named first. A name ending in **-ide** means the compound contains just the two elements named (sodium chlor**ide** = Na and Cl only; the exception is hydroxide, OH^-). A name ending in **-ate** means the negative ion also contains **oxygen** (copper sulf**ate** contains SO_4^{2-}; potassium nitr**ate** contains NO_3^-).

Word and symbol equations

A **word equation** names the reactants and products. A **balanced symbol equation** uses formulae and has the same number of atoms of each element on both sides, because atoms are not created or destroyed in a chemical reaction. You balance an equation by changing the *large numbers in front* of formulae, never the subscripts inside them.

magnesium + hydrochloric acid → magnesium chloride + hydrogen

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

State symbols

Symbol	Meaning	Example
(s)	solid	$CaCO_3(s)$, $Mg(s)$
(l)	liquid (pure liquid, not a solution)	$H_2O(l)$, $Br_2(l)$
(g)	gas	$CO_2(g)$, $O_2(g)$
(aq)	aqueous — dissolved in water	$NaCl(aq)$, $HCl(aq)$

Worked Example: Balance: aluminium + oxygen → aluminium oxide.
Unbalanced: $Al + O_2 \rightarrow Al_2O_3$

Oxygen: 2 on the left, 3 on the right. The lowest common multiple is 6, so use $3O_2$ and $2Al_2O_3$.

Now 4 Al atoms are on the right, so use 4Al.

$4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s)$ Check: Al 4 = 4; O 6 = 6. ✓

Ionic equations (HT)

An **ionic equation** shows only the particles that actually change in a reaction. Ions that are present but do not take part are called **spectator ions** and are left out.

1. Write the balanced symbol equation with state symbols.
2. Split every **aqueous** ionic substance into its ions (solids, liquids, gases and covalent molecules stay as they are).
3. Cross out ions that appear unchanged on both sides.
4. Check the atoms *and* the total charge balance.

Worked Example: Write the ionic equation for the precipitation of silver chloride:

$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

$Ag^+(aq) + NO_3^-(aq) + Na^+(aq) + Cl^-(aq) \rightarrow AgCl(s) + Na^+(aq) + NO_3^-(aq)$

Na^+ and NO_3^- are spectator ions, so: **$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$**

Other ionic equations you will meet: neutralisation, **$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$**; and a displacement such as **$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$**.

Hazard symbols, risks and precautions

Hazard symbols warn about the dangers of chemicals. You should be able to describe what each means and suggest suitable precautions. A **hazard** is something that could cause harm; a **risk** is the chance of harm actually happening. Precautions reduce the risk.

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GHS hazard pictograms (red diamond, black symbol)



Flammable
catches fire easily

Oxidising
helps other things burn

Toxic
can cause death

Corrosive
destroys skin, eyes, metals

Also learn: harmful/irritant (exclamation mark), environmental hazard (dead tree and fish), health hazard, explosive.

Hazard	Example	Typical precaution
Flammable	Ethanol, hydrogen	Keep away from naked flames; use a water bath or electric heater instead of a Bunsen burner
Oxidising	Oxygen, potassium manganate(VII)	Keep away from flammable materials
Toxic	Chlorine, bromine	Use in a fume cupboard; wear gloves
Corrosive	Concentrated acids and alkalis	Wear eye protection and gloves; use the most dilute solution that works
Harmful / irritant	Dilute acids and alkalis, copper sulfate	Wear eye protection; avoid skin contact; wash hands

Exam Tip: In a "risk assessment" question, always pair a specific *hazard* with a specific *precaution* — "wear goggles because the acid is corrosive and could damage the eyes" scores; "be careful" does not.

1.2 Atomic Structure

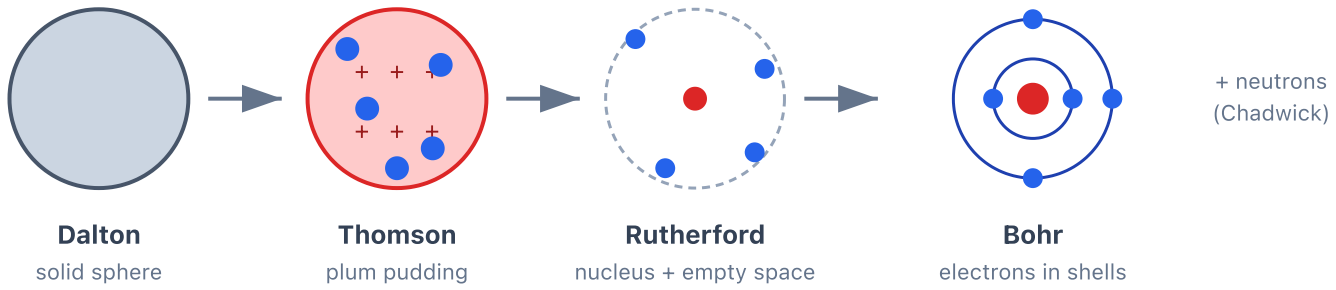
How the model of the atom developed

Our model of the atom has changed as new experimental evidence appeared. Each new model kept what still worked and changed what the evidence disproved.

Scientist / date	Model	Key evidence or idea
Dalton (early 1800s)	Atoms are tiny solid spheres that cannot be divided; atoms of one element are identical; different elements have different atoms	Explained fixed proportions in which elements combine
Thomson (1897)	"Plum pudding": a ball of positive charge with negative electrons embedded in it	Discovery of the **electron** — atoms contain smaller particles, so they are divisible
Rutherford (1909–11)	Nuclear model: tiny, dense, positive **nucleus** containing most of the mass, surrounded by mostly empty space where electrons are	Alpha-particle scattering (Geiger & Marsden): most alpha particles passed straight through gold foil; a few were deflected; a very small number bounced back
Bohr (1913)	Electrons orbit the nucleus in fixed **energy levels** (shells) at specific distances	Explained why atoms emit and absorb light only at certain energies; calculations agreed with observations
Later work (Chadwick, 1932)	Nucleus contains protons *and* neutrons	Discovery of the **neutron**, explaining why atomic masses were larger than the proton count suggested

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Development of the atomic model



Red = positive charge / nucleus; blue = electrons.

> **Exam Tip:** For the alpha-scattering experiment, link each observation to a conclusion: *most passed straight through* → the atom is mostly empty space; *some were deflected* → the nucleus is positively charged; *very few bounced back* → the nucleus is very small and contains most of the mass.

Structure of the atom

An atom has a central **nucleus** containing **protons** and **neutrons**, surrounded by **electrons** arranged in shells (energy levels).

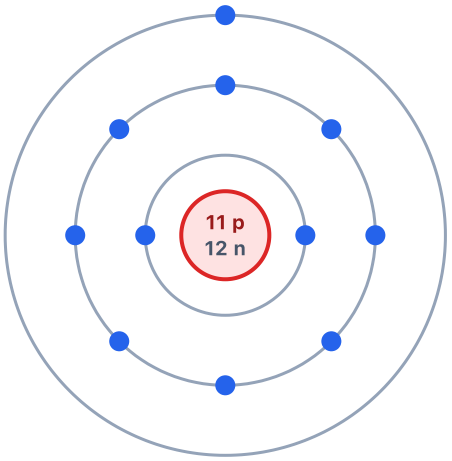
Particle	Relative mass	Relative charge	Location
Proton	1	+1	Nucleus
Neutron	1	0	Nucleus
Electron	1/1836 (negligible)	−1	Shells around the nucleus

- An atom is **neutral** overall because it contains **equal numbers of protons and electrons**; their charges cancel.
- The nucleus is tiny compared with the whole atom: its radius is about **1/100 000** of the atom's radius.
- Almost all of the **mass** of an atom is in the nucleus, because protons and neutrons are far heavier than electrons.

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- Atoms are about **10^{-10} m** (0.1 nm) across, and simple molecules are of the same order of magnitude (about 10^{-10} to 10^{-9} m). The nucleus is about 10^{-14} m across.

A sodium atom, $^{23}_{11}$Na — electronic configuration 2.8.1



Mass number A = 23 (protons + neutrons)

Atomic number Z = 11 (protons)

Neutrons = 23 − 11 = 12

Electrons = 11 (neutral atom)

3 shells → Period 3

1 outer electron → Group 1

Atomic number and mass number

Atomic number (Z) = the number of **protons** in the nucleus. All atoms of the same element have the same number of protons, and this is what identifies the element.
Mass number (A) = the total number of **protons + neutrons** in the nucleus.
Number of neutrons = A − Z.

Atoms are written in the notation $^{A}_{Z}\mathbf{X}$, with the mass number on top and the atomic number below, e.g. $^{12}_{6}$C, $^{35}_{17}$Cl, $^{56}_{26}$Fe.

Worked Example: How many protons, neutrons and electrons are in an atom of $^{39}_{19}$K?
Protons = Z = 19; electrons = 19 (neutral atom); neutrons = 39 − 19 = **20**.

1.3 Isotopes and Relative Atomic Mass

Isotopes are atoms of the **same element** (same number of protons) with **different numbers of neutrons**, and therefore different mass numbers.

Isotopes of an element have the same electronic configuration, so they have the **same chemical properties**. Their physical properties that depend on mass (such as density) differ slightly.

Isotope	Protons	Neutrons	Electrons
^{1}H (hydrogen-1)	1	0	1
^{2}H (deuterium)	1	1	1
^{35}Cl	17	18	17
^{37}Cl	17	20	17

Why relative atomic masses are not whole numbers

Most elements exist naturally as a **mixture of isotopes**. The **relative atomic mass (A_r)** is the **weighted mean** mass of the atoms of an element, taking into account the mass and the abundance of each isotope, compared with 1/12 of the mass of a carbon-12 atom. Because it is an average, it is often not a whole number (e.g. Cl = 35.5, Cu = 63.5).

Calculating relative atomic mass (HT)

$$A_r = \Sigma(\text{isotope mass} \times \% \text{ abundance}) \div 100$$

Worked Example: (HT) Chlorine is 75% ^{35}Cl and 25% ^{37}Cl. Calculate the relative atomic mass of chlorine.

$A_r = (35 \times 75 + 37 \times 25) \div 100 = (2625 + 925) \div 100 = 3550 \div 100 = \mathbf{35.5}$

Worked Example: (HT) Copper consists of 69.2% ^{63}Cu and 30.8% ^{65}Cu. Calculate A_r(Cu) to 1 decimal place.

$A_r = (63 \times 69.2 + 65 \times 30.8) \div 100 = (4359.6 + 2002.0) \div 100 = \mathbf{63.6}$

Exam Tip: Sanity-check your answer: A_r must lie *between* the lightest and heaviest isotope masses and be closer to the more abundant isotope. If abundances are given as ratios (e.g.

> 3 : 1), divide by the total of the ratio (4) instead of 100.

1.4 The Periodic Table

Mendeleev's periodic table

In 1869 **Dmitri Mendeleev** arranged the known elements in order of increasing **relative atomic mass**, but he did not follow that order blindly:

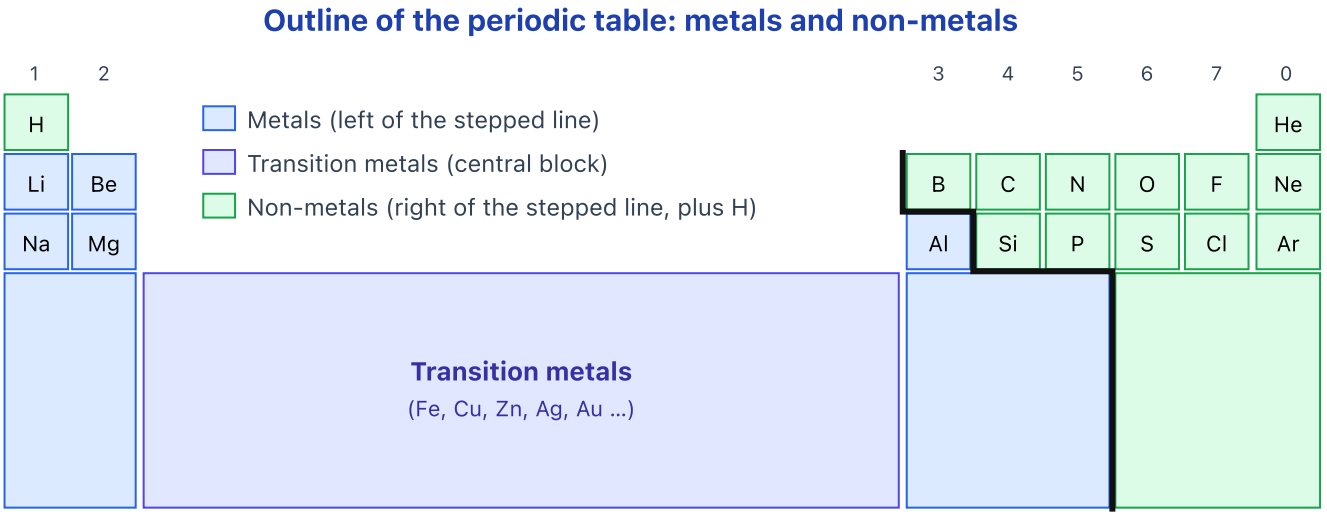
- He placed elements with **similar properties in the same vertical group**, even if this meant swapping the order of some pairs.
- He **left gaps** for elements he believed had not yet been discovered.
- He used the trends in his table to **predict the properties** of these missing elements (e.g. "eka-silicon", later found as germanium, with properties very close to his predictions). These successful predictions convinced other scientists that his table was useful.

Mendeleev swapped some pairs, for example **tellurium (A_r 127.6) and iodine (A_r 126.9)**: ordering by mass would have put iodine in the wrong group, so he put iodine with chlorine and bromine, whose properties it matches. Later, the discovery of **protons** and **isotopes** explained why this was correct: the elements are really in order of **atomic number**, and an element's A_r can be "out of order" because of the relative abundance of its heavier isotopes (tellurium has a high proportion of heavy isotopes).

The modern periodic table

- Elements are arranged in order of increasing **atomic number**.
- The horizontal rows are **periods**; the vertical columns are **groups**.
- Elements in the same group have the **same number of outer-shell electrons** and so similar chemical properties.
- **Metals** are found to the **left** and in the centre; **non-metals** are found to the **right** (top right). Most elements are metals.

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Periods run across (rows); groups run down (columns). Group 0 = noble gases.

1.5 Electronic Configuration

Electrons occupy shells (energy levels) around the nucleus. The lowest-energy shell (closest to the nucleus) fills first. For the first 20 elements:

- 1st shell holds up to **2** electrons
- 2nd shell holds up to **8** electrons
- 3rd shell: for elements 1–20, holds 8 before electrons go into the 4th shell

Element	Z	Configuration	Element	Z	Configuration
H	1	1	Na	11	2.8.1
He	2	2	Mg	12	2.8.2
Li	3	2.1	Al	13	2.8.3
Be	4	2.2	Si	14	2.8.4
B	5	2.3	P	15	2.8.5
C	6	2.4	S	16	2.8.6
N	7	2.5	Cl	17	2.8.7

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O	8	2.6	Ar	18	2.8.8
F	9	2.7	K	19	2.8.8.1
Ne	10	2.8	Ca	20	2.8.8.2

Linking configuration to position in the table

- **Number of occupied shells = period number.** Chlorine (2.8.7) has 3 shells → Period 3.
- **Number of outer-shell electrons = group number** (for Groups 1–7). Chlorine has 7 outer electrons → Group 7.
- Group 0 elements have a **full outer shell** (2 for helium, 8 for the others), which is why they are unreactive.

> **Exam Tip:** You can work out a configuration from the periodic table alone: count across each period. Calcium is the 2nd element in Period 4 → shells 2, 8, 8 then 2 → **2.8.8.2**. Draw configurations as dots or crosses on concentric circles when asked for a diagram.

1.6 Ionic Bonding

Forming ions

> An **ion** is an atom or group of atoms with an overall electric charge, formed when atoms **lose or gain electrons**.
> **Cations** are positive ions (electrons lost). **Anions** are negative ions (electrons gained).

- **Metal** atoms lose their outer-shell electrons to form **positive** ions. Group 1 → 1+, Group 2 → 2+, Group 3 (aluminium) → 3+.
- **Non-metal** atoms gain electrons to fill their outer shell, forming **negative** ions. Group 6 → 2–, Group 7 → 1–.
- The ions formed have the electronic configuration of a **noble gas** (a full outer shell). Na^+ is 2.8, like neon; Cl^- is 2.8.8, like argon.

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Protons, neutrons and electrons in ions

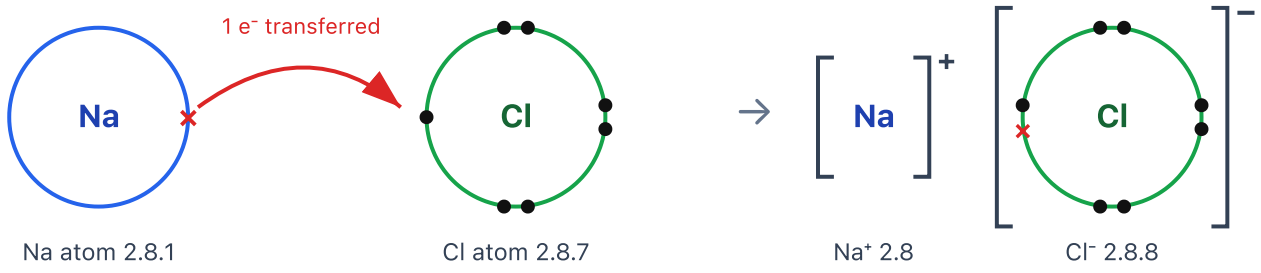
Forming an ion changes only the number of **electrons**; the nucleus is unchanged.

Ion	Protons	Neutrons	Electrons	Reason
$^{23}Na^{+}$	11	12	10	lost 1 electron
$^{24}Mg^{2+}$	12	12	10	lost 2 electrons
$^{16}O^{2-}$	8	8	10	gained 2 electrons
$^{35}Cl^{-}$	17	18	18	gained 1 electron
$^{27}Al^{3+}$	13	14	10	lost 3 electrons

How ionic bonds form

When a metal reacts with a non-metal, **electrons are transferred** from the metal atoms to the non-metal atoms. The oppositely charged ions formed are held together by **strong electrostatic forces of attraction** — this is an **ionic bond**.

Ionic bonding: sodium chloride (dot-and-cross, outer shells shown)



Na atom 2.8.1 — Cl atom 2.8.7 — Na^{+} 2.8 — Cl^{-} 2.8.8

Magnesium oxide: Mg (2.8.2) transfers **2** electrons to O (2.6)
→ Mg^{2+} [2.8] and O^{2-} [2.8]. Both ions have the configuration of neon.
Dots and crosses show which atom the electrons came from — all electrons are identical.

Formulae of ionic compounds

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An ionic compound has no overall charge, so the **total positive charge must equal the total negative charge**.

> **Worked Example:** Work out the formula of aluminium sulfate.
> Ions: Al^{3+} and SO_4^{2-}. Lowest common multiple of 3 and 2 is 6 → 2 × Al^{3+} (+6) and 3 × SO_4^{2-} (−6).
> Formula: **$Al_2(SO_4)_3$** (brackets are needed around a group ion when there is more than one of it).

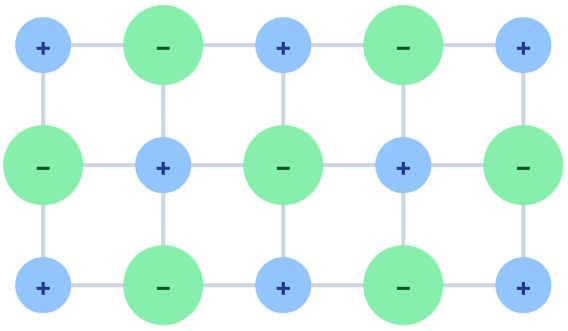
Compound	Ions	Formula
Calcium chloride	Ca^{2+}, Cl^-	$CaCl_2$
Sodium oxide	Na^+, O^{2-}	Na_2O
Ammonium nitrate	NH_4^+, NO_3^-	NH_4NO_3
Calcium hydroxide	Ca^{2+}, OH^-	$Ca(OH)_2$
Iron(III) oxide	Fe^{3+}, O^{2-}	Fe_2O_3
Potassium carbonate	K^+, CO_3^{2-}	K_2CO_3

Structure of ionic compounds

Ionic compounds form **giant ionic lattices**: a regular three-dimensional arrangement of alternating positive and negative ions, held together by strong electrostatic forces acting in **all directions**.

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Giant ionic lattice (sodium chloride, one layer)



- Na^+ (smaller cation)
- Cl^- (larger anion)

Each ion is surrounded by ions of opposite charge.

Strong electrostatic attraction in all directions → giant structure.

1.7 Covalent Bonding

A **covalent bond** is a **shared pair of electrons** between two atoms. Covalent bonds form between **non-metal atoms**. The bond is strong because both nuclei are attracted to the shared pair.

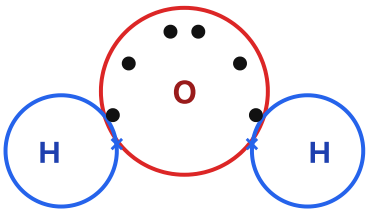
By sharing electrons, each atom usually achieves a full outer shell. A **double bond** (two shared pairs) is found in O_2 and CO_2; a **triple bond** (three shared pairs) is found in N_2.

Simple molecules you must be able to draw

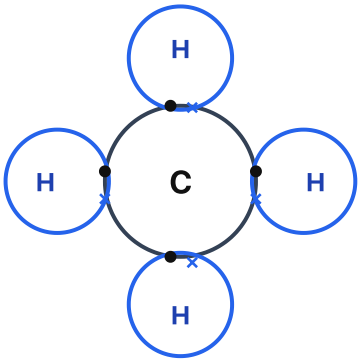
Molecule	Bonds	Notes
Hydrogen, H_2	1 single bond (H–H)	Each H has 2 electrons in its shell
Hydrogen chloride, HCl	1 single bond	Cl also has 3 non-bonding (lone) pairs
Water, H_2O	2 single bonds	O has 2 lone pairs
Methane, CH_4	4 single bonds	C has no lone pairs
Oxygen, O_2	1 double bond (O=O)	Each O has 2 lone pairs
Carbon dioxide, CO_2	2 double bonds (O=C=O)	Each O has 2 lone pairs

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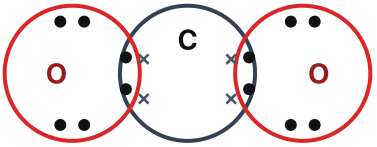
Dot-and-cross diagrams (outer shells only)



Water, H_2O



Methane, CH_4



Carbon dioxide, CO_2
two C=O double bonds

> **Exam Tip:** When drawing dot-and-cross diagrams, put shared pairs in the overlap of the circles and count: every atom (except H, which needs 2) should end with 8 outer electrons. Check the total number of electrons equals the sum of the atoms' outer electrons.

1.8 Structure, Bonding and Properties

Substances can be classified by their structure and bonding into **ionic**, **simple molecular (covalent)**, **giant covalent** and **metallic**. Polymers are a further important group of covalent substances. The type of structure explains the physical properties.

Ionic compounds

- **High melting and boiling points** — a lot of energy is needed to overcome the many strong electrostatic forces between oppositely charged ions throughout the lattice.
- **Do not conduct electricity when solid** — ions are held in fixed positions and cannot move.
- **Conduct when molten or dissolved in water** — the ions are free to move and carry charge.
- Many are **soluble in water**.

Simple molecular substances

Examples: H_2, O_2, Cl_2, H_2O, CO_2, CH_4. Atoms *within* each molecule are held by strong covalent bonds, but the forces *between* molecules (**intermolecular forces**) are **weak**.

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- **Low melting and boiling points** — only the weak intermolecular forces need to be overcome; the covalent bonds do not break when the substance melts or boils. Many are gases or liquids at room temperature.
- Intermolecular forces get stronger as molecules get larger, so melting and boiling points generally **increase with molecule size**.
- **Do not conduct electricity** — the molecules have no overall charge and there are no free electrons or ions.

> **Exam Tip:** The most common lost mark in this topic: saying covalent bonds break when a simple molecular substance boils. They do not — only the *intermolecular forces* are overcome. Always name which forces are involved.

Giant covalent substances

Very large numbers of atoms joined by covalent bonds in a continuous network, e.g. **diamond**, **graphite** and **silicon dioxide (silica)**. They have **very high melting points** because many strong covalent bonds must be broken, and they are insoluble in water.

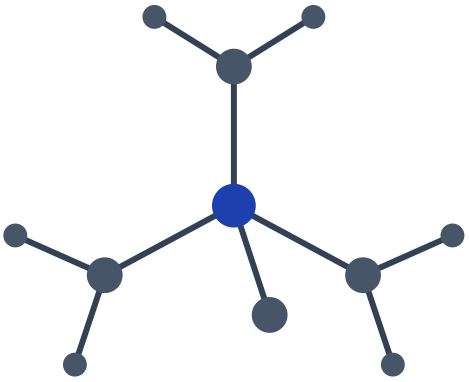
Allotropes of carbon

Form	Structure	Properties	Uses
Diamond	Giant covalent; each C bonded to **4** others in a rigid tetrahedral network	Very hard; very high melting point; does **not** conduct electricity (no delocalised electrons)	Cutting tools, drill tips
Graphite	Giant covalent; each C bonded to **3** others in flat hexagonal **layers**; weak forces between layers; one electron per atom **delocalised**	Soft and slippery (layers slide); very high melting point; **conducts** electricity (delocalised electrons move along the layers)	Electrodes, lubricant, pencil "lead"

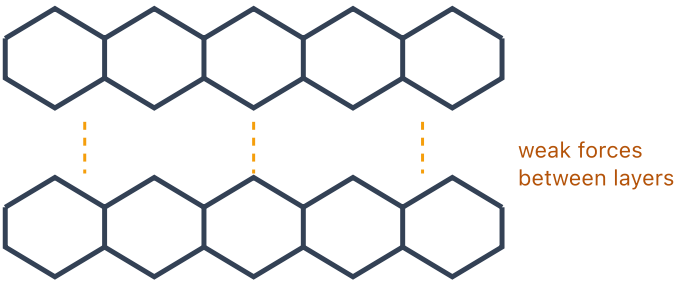
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Graphene	A **single layer** of graphite (one atom thick)	Very strong; conducts electricity well; very light; almost transparent	Composites, electronics, touchscreens
Fullerenes	Molecules of carbon atoms in hollow shapes, mainly hexagonal rings (with some 5- or 7-membered rings). Buckminsterfullerene is **C_{60}** (spherical). **Nanotubes** are cylindrical fullerenes	C_{60}: simple molecular, so lower melting point than diamond and graphite, slippery; nanotubes: very high length-to-diameter ratio, high tensile strength, conduct electricity	Drug delivery (hollow cage), lubricants, catalysts (large surface area), reinforcing materials (nanotubes)

Diamond and graphite



Diamond: each C forms 4 bonds



**Graphite: each C forms 3 bonds;
1 delocalised electron per atom**

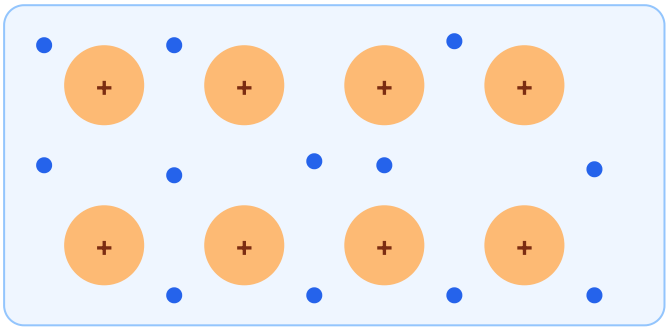
Polymers

Polymers such as poly(ethene) are very large molecules made of many repeating units joined by strong covalent bonds into long chains. Because the molecules are so large, the intermolecular forces between chains are relatively strong, so polymers are **solids at room temperature**. Most do not conduct electricity.

Metallic bonding

Metallic bonding is the electrostatic attraction between a lattice of **positive metal ions** and a "sea" of **delocalised electrons** (the outer-shell electrons, which are free to move through the whole structure).

Metallic bonding: positive ions in a sea of delocalised electrons



- Regular layers of positive ions
- Delocalised electrons move freely

→ conduct electricity and heat

- Layers slide over one another

→ malleable and ductile

- **High melting points** (mostly) — strong attraction between the positive ions and delocalised electrons.
- **Good conductors of electricity and heat** — delocalised electrons carry charge and energy through the structure.
- **Malleable** (can be hammered into shape) — the layers of ions can slide over each other while the delocalised electrons keep holding them together.
- Typical metal properties: shiny, high density, solid at room temperature (except mercury), form positive ions and basic oxides. Typical non-metal properties: dull, brittle, low melting points (except giant covalent ones), poor conductors (except graphite), form negative ions and acidic oxides.

Summary of structure types

Structure	Particles and bonding	Melting point	Conducts?	Example
Ionic	Ions; electrostatic attraction	High	Only when molten or dissolved	NaCl, MgO
Simple molecular	Small molecules; covalent bonds within,	Low	No	H_2O, CO_2, CH_4

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	weak intermolecular forces between			
Giant covalent	Atoms; network of covalent bonds	Very high	No (except graphite, graphene)	Diamond, SiO_2
Metallic	Positive ions + delocalised electrons	Generally high	Yes, solid and liquid	Cu, Fe, Al
Polymer	Very long covalent chains	Solid at room temperature; soften over a range	No	Poly(ethene)

Worked Example: A substance melts at 801 °C. It does not conduct as a solid but does conduct when molten. Identify its structure.
High melting point → strong forces throughout a giant structure. Conducts only when molten → charged particles that become mobile on melting → **giant ionic lattice** (it is sodium chloride).

Limitations of models

Chemists use several models to represent bonding and structure. Each shows something useful but has limitations:

Model	Useful for	Limitations
Dot-and-cross diagram	Showing which atom electrons came from and how many are shared/transferred	Does not show the 3D shape or relative sizes of atoms; suggests electrons from different atoms are different
2D structural formula (e.g. H–O–H)	Showing which atoms are bonded and how many bonds	Does not show shape, lone pairs or bond angles

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Ball-and-stick model	Showing 3D arrangement and bond angles	Bonds are shown as sticks, but there is no physical "stick"; atoms are spaced too far apart and sizes not to scale
Space-filling model	Showing relative sizes and overall shape	Hard to see bonds or the structure inside
Lattice diagram	Showing the regular repeating arrangement of ions	Only shows a tiny part of the structure; does not show forces or movement

1.9 Calculations: Formula Mass, Empirical Formulae and Concentration

Relative formula mass

Relative formula mass (M_r) is the sum of the relative atomic masses of all the atoms in the formula.

Worked Example: Calculate M_r of calcium hydroxide, $Ca(OH)_2$. (Ca = 40, O = 16, H = 1)
$M_r = 40 + 2 \times (16 + 1) = 40 + 34 =$ **74**

Worked Example: Calculate M_r of hydrated copper sulfate, $CuSO_4 \cdot 5H_2O$. (Cu = 63.5, S = 32, O = 16, H = 1)
$CuSO_4 = 63.5 + 32 + 64 = 159.5$; $5H_2O = 5 \times 18 = 90 \rightarrow M_r =$ **249.5**

Percentage by mass of an element in a compound:

$$\% \text{ by mass} = (A_r \times \text{number of atoms of that element}) \div M_r \times 100$$

Empirical and molecular formulae

> The **empirical formula** is the **simplest whole-number ratio** of atoms of each element in a compound. The **molecular formula** is the actual number of atoms of each element in one molecule.

Examples: ethane C_2H_6 has empirical formula CH_3; glucose $C_6H_{12}O_6$ has empirical formula CH_2O. For ionic and giant covalent substances we always use the empirical formula (NaCl, SiO_2).

To find an empirical formula from masses (or percentages):

1. Write down the mass (or %) of each element.
2. Divide each by its A_r.
3. Divide all answers by the smallest one.
4. If necessary, multiply to get whole numbers (e.g. 1.5 → ×2).

> **Worked Example:** A compound contains 2.4 g of carbon and 0.6 g of hydrogen. Its M_r is 30. Find the empirical and molecular formulae.
> C: 2.4 ÷ 12 = 0.2 H: 0.6 ÷ 1 = 0.6 → ratio 0.2 : 0.6 = 1 : 3 → empirical formula **CH_3**.
> M_r of CH_3 = 15; 30 ÷ 15 = 2 → molecular formula **C_2H_6**.

Practical: finding the empirical formula of magnesium oxide

1. Weigh a clean, dry crucible and lid. Add a coiled strip of magnesium ribbon (cleaned with sandpaper to remove the oxide layer) and weigh again.
2. Heat strongly on a pipeclay triangle. **Lift the lid slightly** from time to time with tongs to let oxygen in, while preventing white magnesium oxide smoke escaping.
3. When the magnesium no longer glows, remove the lid and heat for a few more minutes. Allow to cool and reweigh.
4. Repeat heating, cooling and weighing until the mass is constant (**heating to constant mass**) to ensure all the magnesium has reacted.

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Worked Example: Mass of Mg = 0.24 g; mass of magnesium oxide = 0.40 g. Find the empirical formula. (Mg = 24, O = 16)
Mass of O = 0.40 − 0.24 = 0.16 g.
Mg: 0.24 ÷ 24 = 0.010 O: 0.16 ÷ 16 = 0.010 → ratio 1 : 1 → **MgO**.

Exam Tip: Errors in this practical: if MgO smoke escapes, the final mass is too low; if some Mg does not react, the oxygen mass is too low. Both make the ratio of O to Mg smaller than it should be.

Conservation of mass

The **law of conservation of mass**: no atoms are lost or made during a chemical reaction, so the total mass of products equals the total mass of reactants.

- In a **closed system** (nothing can enter or leave, e.g. a precipitation reaction in a sealed flask), the mass measured on a balance stays the same.
- In a **non-enclosed system** the mass on the balance may appear to change:
  - **Increase** if a reactant comes from the air — e.g. when magnesium burns, oxygen from the air is added, so the solid gains mass.
  - **Decrease** if a gaseous product escapes — e.g. thermal decomposition of a metal carbonate ($CaCO_3 \rightarrow CaO + CO_2$), or a carbonate reacting with an acid, loses CO_2 to the air.

Worked Example: 12.5 g of calcium carbonate is heated until it fully decomposes, leaving 7.0 g of calcium oxide. What mass of carbon dioxide was produced?
By conservation of mass: 12.5 − 7.0 = **5.5 g** of CO_2 escaped.

Reacting masses from a balanced equation

The balanced equation tells you the ratio in which substances react. Multiply each M_r by its balancing number to get the reacting masses, then scale to the mass you are given. (Higher-tier students can also do this with moles — see 1.10.)

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Worked Example: What mass of magnesium oxide forms when 6.0 g of magnesium burns completely? $2Mg + O_2 \rightarrow 2MgO$ (Mg = 24, O = 16)
From the equation: 2 × 24 = 48 g of Mg gives 2 × 40 = 80 g of MgO.
So 6.0 g of Mg gives 80 × (6.0 ÷ 48) = **10 g** of MgO.

Concentration in g dm^{-3}

$$\text{concentration (g dm}^{-3}\text{)} = \text{mass of solute (g)} \div \text{volume of solution (dm}^3\text{)}$$

1 dm³ = 1000 cm³, so divide a volume in cm³ by 1000 to convert to dm³.

Worked Example: 5.0 g of sodium hydroxide is dissolved in water to make 250 cm³ of solution. Calculate the concentration.
Volume = 250 ÷ 1000 = 0.250 dm³. Concentration = 5.0 ÷ 0.250 = **20 g dm^{-3}**.

Worked Example: What mass of solute is in 40 cm³ of a 15 g dm^{-3} solution?
Mass = concentration × volume = 15 × 0.040 = **0.60 g**.

1.10 The Mole (HT)

One **mole** of a substance contains the same number of particles (atoms, molecules, ions or electrons) as there are atoms in 12 g of carbon-12. This number is the **Avogadro constant**: **6.02 × 10^{23} per mole**.
The mass of one mole of a substance in grams is numerically equal to its A_r or M_r (e.g. 1 mol of H_2O has a mass of 18 g).

$$\text{number of moles} = \text{mass (g)} \div M_r \quad \leftrightarrow \quad n = m \div M_r$$

$$\text{number of particles} = \text{moles} \times 6.02 \times 10^{23}$$

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Worked Example: How many molecules are in 8.8 g of carbon dioxide? (C = 12, O = 16)
$M_r(CO_2) = 44$ → moles = 8.8 ÷ 44 = 0.20 mol.
Molecules = 0.20 × 6.02 × 10^{23} = **1.2 × 10^{23}** (2 s.f.). Number of *atoms* = 3 × that = 3.6 × 10^{23}.

Reacting masses using moles

1. Calculate the moles of the substance whose mass you know ($n = m \div M_r$).
2. Use the **mole ratio** from the balanced equation to find moles of the substance you want.
3. Convert back to mass ($m = n \times M_r$).

Worked Example: What mass of iron is produced from 32 g of iron(III) oxide? $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ (Fe = 56, O = 16)
$M_r(Fe_2O_3) = 160$ → moles = 32 ÷ 160 = 0.20 mol.
Ratio Fe_2O_3 : Fe = 1 : 2 → moles Fe = 0.40 mol.
Mass Fe = 0.40 × 56 = **22.4 g**.

Deducing an equation from reacting masses

If you know the masses of all reactants and products, convert each to moles and then find the simplest whole-number ratio. These numbers are the balancing numbers in the equation.

Worked Example: 4.8 g of magnesium reacts with 3.2 g of oxygen to form 8.0 g of magnesium oxide. Deduce the balanced equation. (Mg = 24, O = 16)
Mg: 4.8 ÷ 24 = 0.20 mol; O_2: 3.2 ÷ 32 = 0.10 mol; MgO: 8.0 ÷ 40 = 0.20 mol.
Ratio 0.20 : 0.10 : 0.20 = 2 : 1 : 2 → **$2Mg + O_2 \rightarrow 2MgO$**.

Limiting reactants

The **limiting reactant** is the reactant that is completely used up; it limits the amount of product that can form. The other reactant is **in excess** — some is left over at the end.

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The mass of product formed is **directly proportional** to the amount of the limiting reactant: double the limiting reactant and you double the product (as long as the other reactant stays in excess). Chemists often use an excess of one (cheaper) reactant to make sure all of a more expensive reactant reacts.

Worked Example: 6.0 g of magnesium is added to a solution containing 0.40 mol of HCl. $Mg + 2HCl \rightarrow MgCl_2 + H_2$. Which reactant is limiting, and what mass of hydrogen forms? (Mg = 24, H = 1)
Moles Mg = 6.0 ÷ 24 = 0.25 mol. To react fully it would need 2 × 0.25 = 0.50 mol HCl, but only 0.40 mol is available → **HCl is limiting**, Mg is in excess.
Moles H_2 = 0.40 ÷ 2 = 0.20 mol → mass = 0.20 × 2 = **0.40 g**.

Exam Tip: To find the limiting reactant, divide the moles of each reactant by its balancing number in the equation; the *smallest* result is the limiting reactant. Always base the product calculation on the limiting reactant.

Key Points Summary

- Recall formulae of common elements, compounds and ions; -ide = two elements, -ate = contains oxygen.
- Balance equations by changing the numbers in front of formulae; include state symbols (s), (l), (g), (aq).
- **(HT)** Ionic equations leave out spectator ions — e.g. $H^+ + OH^- \rightarrow H_2O$.
- Hazard symbols: flammable, oxidising, toxic, corrosive, harmful; match each hazard to a precaution.
- Atomic model: Dalton → Thomson (electron) → Rutherford (nucleus, alpha scattering) → Bohr (shells) → neutron (Chadwick).
- Proton: mass 1, charge +1. Neutron: mass 1, charge 0. Electron: mass 1/1836, charge –1. Atoms ~10^{-10} m; nucleus ~1/100 000 of atomic radius but contains almost all the mass.
- Atomic number = protons; mass number = protons + neutrons. Isotopes: same protons, different neutrons.

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- A_r = weighted mean of isotope masses, so often not a whole number; **(HT)** calculate it from isotope masses and abundances.
- Mendeleev ordered by atomic mass, left gaps, predicted properties and swapped pairs such as Te/I; the modern table is ordered by atomic number.
- Electronic configuration (first 20 elements): shells = period; outer electrons = group.
- Ionic bonding: electron transfer from metal to non-metal; giant lattice; high mp; conducts when molten or aqueous.
- Covalent bonding: shared pairs between non-metals; simple molecules have weak intermolecular forces and low mp; giant covalent structures have very high mp.
- Diamond (4 bonds, hard, non-conductor), graphite (3 bonds, layers, conducts), graphene (single layer), fullerenes (C_{60}, nanotubes).
- Metallic bonding: positive ions + delocalised electrons → conductors, malleable.
- Every model of bonding has limitations — know at least one for each.
- M_r = sum of A_r; empirical formula = simplest whole-number ratio (MgO practical: heat to constant mass).
- Mass is conserved; apparent changes in open systems come from gases entering or leaving.
- Reacting masses: scale the masses given by (M_r × balancing number) in the equation.
- Concentration (g dm^{-3}) = mass ÷ volume in dm^3.
- **(HT)** Moles = mass ÷ M_r; Avogadro constant 6.02×10^{23}; use mole ratios for reacting masses, deducing equations and limiting reactants.

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